

LECTURE NOTES MATH255

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1. GALOIS GROUPS FOR POLYNOMIALS OF LOW DEGREE

In these notes we give some tools to calculate Galois groups of the splitting field E_f of a polynomial $f(x) \in \mathbb{Q}[x]$ of low degree. It will be useful to remember the interpretation of the Galois group that we used during the last class. If $f(x) \in \mathbb{Q}[x]$ has zeros $\alpha_1, \dots, \alpha_n$ then any $\psi \in \text{Gal}(E_f/\mathbb{Q})$ defines a permutation $\sigma_\psi : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ by $\psi(\alpha_j) = \alpha_{\sigma_\psi(j)}$. With this observation we can view $\text{Gal}(E_f/\mathbb{Q}) \leq S_n$ as a subgroup of the symmetric group.

1.1. Polynomials of degree 2. We will

Example 1.1. Let $f(x) = x^2 + 4x + 1$. We calculate the zeros of $f(x)$ as:

$$(1.1) \quad \alpha_1 = -2 + \sqrt{\frac{16}{4} - 1} = -2 + \sqrt{3},$$

$$(1.2) \quad \alpha_2 = -2 - \sqrt{3}.$$

We note that any $\psi \in \text{Gal}(E_f/\mathbb{Q})$ maps one zero to another, so there are only two possibilities for ψ :

$$(1.3) \quad \psi(\alpha_1) = \alpha_2, \quad \psi(\alpha_2) = \psi(\alpha_1),$$

$$(1.4) \quad \psi(\alpha_1) = \alpha_1, \quad \psi(\alpha_2) = \alpha_2.$$

The second map is the identity, so is always in the Galois group. The first map is also in the Galois group since $f(x)$ is irreducible. Indeed, irreducibility of $f(x)$ implies that there is some $\psi \in \text{Gal}(E_f/\mathbb{Q})$ that maps α_1 to any conjugate of α_1 , in particular there must be some $\psi \in \text{Gal}(E_f/\mathbb{Q})$. We conclude that $\text{Gal}(E_f/\mathbb{Q}) \cong S_2 \cong \mathbb{Z}_2$.

Example 1.2. Let $f(x) = x^2 - 3x + 2$. Note that $f(1) = 1 - 3 + 2 = 0$ and $f(2) = 4 - 6 + 2 = 0$. That is, $f(x)$ has only rational zeros so $E_f = \mathbb{Q}$ and so $\text{Gal}(E_f/\mathbb{Q}) = \{\text{id}\}$ is the trivial group.

These two examples highlights the only two possibilities for Galois groups of splitting fields of polynomials of degree 2: Either the zeros are not rational, in which case $\text{Gal}(E_f/\mathbb{Q}) \cong S_2 \cong \mathbb{Z}_2$, or the zeros are rational in which case the Galois group is trivial.

Problem 1. Find the Galois group for $f(x) = x^2 - 3x + 1$.

Problem 2. Find the Galois groups for $f(x) = x^2 - 4x + 4$.

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1.2. Polynomials of degree 3.

Example 1.3. Let $f(x) = x^3 + 3x^2 - 3x - 1$. Note that $f(1) = 1 + 3 - 3 - 1 = 0$, and we can write $f(x) = (x^2 + 4x + 1)(x - 1)$. It follows that the zeros of $f(x)$ are $1, \alpha_1, \alpha_2$ where α_1, α_2 are the zeros from Example 1.1. We then obtain the splitting field for $f(x)$ as:

$$(1.5) \quad E_f = \mathbb{Q}(\alpha_1, \alpha_2, 1) = \mathbb{Q}(\alpha_1, \alpha_2).$$

Note that the splitting field is the same as the splitting field in Example 1.1, so the Galois group (which only depends on the splitting field!) is also the same. We conclude that $\text{Gal}(E_f/\mathbb{Q}) \cong S_2 \cong \mathbb{Z}_2$. If we want to view this as a subgroup of S_3 , since $f(x)$ has degree 3, it corresponds to the subgroup $H \leq S_3$ where $\sigma \in H$ satisfies $\sigma(3) = 3$ (here we consider 1 as the third zero, and 1 is fixed by any element of $\text{Gal}(E_f/\mathbb{Q})$ since 1 is rational).

Example 1.4. Consider the polynomial $f(x) = x^3 + 5x + 5$. We note that $f(x)$ is irreducible by Eisenstein's criterion since 1 is not divisible by 5, 5 is divisible by 5, but the constant term is not divisible by $5^2 = 25$. A proof that is completely analogous to the proof for degree 5 polynomials in the notes, shows that $\text{Gal}(E_f/\mathbb{Q})$ must contain a 3-cycle, since $f(x)$ is irreducible. We claim that $\text{Gal}(E_f/\mathbb{Q})$ also contains a transposition. We will proceed as in the notes, and show that $f(x)$ has precisely 1 real zero. To see this, note that $f'(x) = 3x^2 + 5 > 0$ for all x . That is, $f(x)$ is strictly increasing which implies that $f(x)$ can have at most 1 zero. On the other hand, $f(x)$ has odd degree with leading coefficient > 0 so :

$$(1.6) \quad f(x) \rightarrow -\infty \text{ as } x \rightarrow -\infty,$$

$$(1.7) \quad f(x) \rightarrow \infty \text{ as } x \rightarrow \infty.$$

By the intermediate value theorem we know that $f(x)$ has at least one real zero, so $f(x)$ has precisely one real zero. We now obtain a transposition in $\text{Gal}(E_f/\mathbb{Q})$ by complex conjugation. Since $\text{Gal}(E_f/\mathbb{Q})$ contains a 3-cycle we know that 3 divides $|\text{Gal}(E_f/\mathbb{Q})|$, and since $\text{Gal}(E_f/\mathbb{Q})$ contains a transposition we know that 2 divides $|\text{Gal}(E_f/\mathbb{Q})|$. We conclude that $|\text{Gal}(E_f/\mathbb{Q})| \geq 6$. On the other hand we have $\text{Gal}(E_f/\mathbb{Q}) \leq S_3$ and $|S_3| = 6$ so we conclude that $\text{Gal}(E_f/\mathbb{Q}) \cong S_3$.

Example 1.5. As we used in the previous example, if $f(x) \in \mathbb{Q}[x]$ is irreducible over $\mathbb{Q}$ and has degree 3 then $\text{Gal}(E_f/\mathbb{Q})$ contains a 3-cycle. So we always, in this case, have $|\text{Gal}(E_f/\mathbb{Q})| \geq 3$. Since $|S_3| = 6$ and since $|\text{Gal}(E_f/\mathbb{Q})|$ divides $|S_3|$ by Lagrange's theorem, this only leaves two possibilities for $\text{Gal}(E_f/\mathbb{Q})$: its either isomorphic to A_3 (in which case $|\text{Gal}(E_f/\mathbb{Q})| = 3$), or its isomorphic to S_3 as in the previous example. The second case is characterized by $\text{Gal}(E_f/\mathbb{Q})$ containing a transposition. An explicit polynomial that has Galois group A_3 is $f(x) = x^3 - 3x + 1$, however the standard proof of this fact uses something we have not covered in the course, the *discriminant* of $f(x)$.

For degree 3 polynomials we have the following possibilities for Galois groups of splitting fields for $f(x) \in \mathbb{Q}[x]$:

- (i) if $f(x)$ is not irreducible, then the calculation of the Galois group can be reduced to the calculation of the Galois group of a degree 2 polynomial,
- (ii) if $f(x)$ is irreducible, then we either have $\text{Gal}(E_f/\mathbb{Q}) \cong S_3$ or $\text{Gal}(E_f/\mathbb{Q}) \cong A_3$, the former case being characterized by $\text{Gal}(E_f/\mathbb{Q})$ containing a transposition,
- (iii) if $f(x)$ is irreducible with 1 real zero and 2 non-real zeros then $\text{Gal}(E_f/\mathbb{Q}) \cong S_3$ since $\text{Gal}(E_f/\mathbb{Q})$ contains a transposition.

Problem 3. Let $f(x) = x^3 - 6x^2 + 11x - 6$, calculate $\text{Gal}(E_f/\mathbb{Q})$.

Problem 4. Let $f(x) = x^3 + 10x - 5$, calculate $\text{Gal}(E_f/\mathbb{Q})$.

Problem 5. Let $f(x) = x^3 - 3x + 5$, calculate $\text{Gal}(E_f/\mathbb{Q})$.

1.3. Higher degree polynomials. Calculation of Galois groups of high degree gets more difficult the higher the degree in general. However, even though calculation for general polynomials might be difficult, there are sufficient conditions that allow for computations. For example, we have the following quite general criteria that is sometimes useful:

Lemma 1.1. *Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of degree p with p prime. If $f(x)$ has $p - 2$ real zeros and 2 non-real zeros then $\text{Gal}(E_f/\mathbb{Q}) \cong S_p$.*

This lemma generalizes what we used in the lecture notes, and also the example for degree 3 polynomials.

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